



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

- **Objective:** predict the successful landing of the SpaceX Falcon 9 first-stage booster to support cost-efficient mission planning.
- **Data Preparation:** launch and landing data were collected from public SpaceX APIs and cleaned, transformed, and formatted for analysis.
- **Exploratory Analysis:** EDA was conducted to examine variable distributions, trends, and relationships affecting landing outcomes.
- **Visualization:** interactive visualizations were used to highlight key patterns and correlations in the data.
- **Modeling:** multiple supervised machine learning classifiers (e.g., classification tree, linear regression) were trained and evaluated using standard performance metrics.
- **Key Findings:** several launch-related features (e.g., launch site, payload mass) show strong correlation with landing success.
- **Conclusion:** the decision tree classifier delivered the best balance of predictive performance and interpretability.

Introduction

- SpaceX's Falcon 9 program enables reusable rocket technology, significantly reducing launch costs.
- Successful first-stage booster landings are essential to achieving reusability and operational efficiency.
- Predicting landing outcomes can support mission planning and risk mitigation.
- This project uses historical Falcon 9 launch data and machine learning models to predict first-stage landing success.



By Joel Kowsky - <https://www.flickr.com/photos/nasahqphoto/49953835192/in/photostream/>

Section 1

Methodology

Methodology

- **Data Collection:** launch and landing data were gathered from public SpaceX APIs and online datasets (i.e., webscrapping).
- **Data Wrangling and Processing:** data was cleaned, filtered, and transformed (using Pandas and Numpy) to ensure consistency and usability.
- **Exploratory Data Analysis:** SQL and Pandas combined with Visualization (e.g., Matplotlib and Seaborn) were used to analyze trends and feature relationships.
- **Interactive Analytics:** Folium maps and Plotly Dash dashboards enabled interactive data exploration of the data.
- **Predictive Modeling:** classification models (e.g, SVM, Classification Trees, and Logistic Regression) were built, tuned, and evaluated to predict landing success.

Data Collection



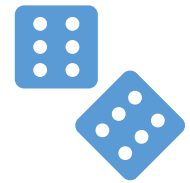
Objective: predict Falcon 9 first-stage landing success.



Data Sources: public SpaceX APIs and open datasets.



Features Collected: launch parameters, payload mass, orbit type, launch site, landing outcome.



Integration: datasets merged using unique flight identifiers and cleaned.

Data Collection – SpaceX API

- **API:** data retrieved using HTTP GET requests to SpaceX public REST endpoints (e.g., for past launches: <https://api.spacexdata.com/v4/launches/past>)
- **Endpoints Used:** launches, rockets, payloads, cores, and landing outcomes.
- **Response Format:** JSON responses parsed into structured tabular data using PANDAS.
- **Data formatting:** filtered records to include only Falcon 9 launches based on booster version
- **Data cleaning:** dropped rows with missing launch dates
- **Data correction:** missing payload mass values filled using the mean payload mass value (column).

FlightNumber		Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad		Block	ReusedCount	Serial	Longitude	Latitude
4	1	2010-06-04	Falcon 9	6187.245062	LEO	CCSFS SLC 40	None None	1	False	False	False	None		1	0	B0003	-80.577366	28.561857
5	2	2012-05-22	Falcon 9	525.000000	LEO	CCSFS SLC 40	None None	1	False	False	False	None		1	0	B0005	-80.577366	28.561857
6	3	2013-03-01	Falcon 9	677.000000	ISS	CCSFS SLC 40	None None	1	False	False	False	None		1	0	B0007	-80.577366	28.561857
7	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	None		1	0	B1003	-120.610829	34.632093
8	5	2013-12-03	Falcon 9	3170.000000	GTO	CCSFS SLC 40	None None	1	False	False	False	None		1	0	B1004	-80.577366	28.561857
...	...	...	...	...	...	...	...	...	...	...	...	...		...	...	...	...	...
89	82	2020-09-03	Falcon 9	15600.000000	VLEO	KSC LC 39A	True ASDS	2	True	True	True	5e9e3032383ecb6bb234e7ca	5	12	B1060	-80.603956	28.608058	
90	83	2020-10-06	Falcon 9	15600.000000	VLEO	KSC LC 39A	True ASDS	3	True	True	True	5e9e3032383ecb6bb234e7ca	5	13	B1058	-80.603956	28.608058	
91	84	2020-10-18	Falcon 9	15600.000000	VLEO	KSC LC 39A	True ASDS	6	True	True	True	5e9e3032383ecb6bb234e7ca	5	12	B1051	-80.603956	28.608058	
92	85	2020-10-24	Falcon 9	15600.000000	VLEO	CCSFS SLC 40	True ASDS	3	True	True	True	5e9e3033383ecbb9e534e7cc	5	12	B1060	-80.577366	28.561857	
93	86	2020-11-05	Falcon 9	3681.000000	MEO	CCSFS SLC 40	True ASDS	1	True	False	True	5e9e3032383ecb6bb234e7ca	5	8	B1062	-80.577366	28.561857	
86 rows × 17 columns																		

Data Collection - Scraping

- **Source:** Falcon 9 historical launch data ([List of Falcon 9 and Falcon Heavy launches – Wikipedia](#))
- **Data access:** web pages retrieved using HTTP GET requests
- **HTML parsing:** page structure inspected and parsed with BeautifulSoup.
- **Data extraction and structuring:** several features were extracted from the HTML table (e.g., launch date, booster version, payload, orbit, etc) and were converted into a Pandas Dataframe.

Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version	Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.07B0003.18	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0004.18	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.18	No attempt\n	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.07B0006.18	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.07B0007.18	No attempt\n	1 March 2013	15:10
...	...	...	...	...	...	...	...	...	...	...	...
116	117	CCSFS	Starlink	15,600 kg	LEO	SpaceX	Success\n	F9 B5B1051.10657	Success	9 May 2021	06:42
117	118	KSC	Starlink	~14,000 kg	LEO	SpaceX	Success\n	F9 B5B1058.8660	Success	15 May 2021	22:56
118	119	CCSFS	Starlink	15,600 kg	LEO	SpaceX	Success\n	F9 B5B1063.2665	Success	26 May 2021	18:59
119	120	KSC	SpaceX CRS-22	3,328 kg	LEO	NASA	Success\n	F9 B5B1067.1668	Success	3 June 2021	17:29
120	121	CCSFS	SXM-8	7,000 kg	GTO	Sirius XM	Success\n	F9 B5	Success	6 June 2021	04:26
121 rows × 11 columns											

Data Wrangling

- **EDA & Outcome Analysis**

- Explored launch outcomes, landing types, payload mass, orbits, and sites (e.g., number and occurrence of each orbit).
- Analyzed landing outcome categories: True / False Ocean, True / False RTLS, True / False ASDS.

- **Label Definition**

- Landing outcomes mapped to binary labels: 1 = Successful landing 0 = Unsuccessful landing
- Created column 'Class' based on landing outcome.



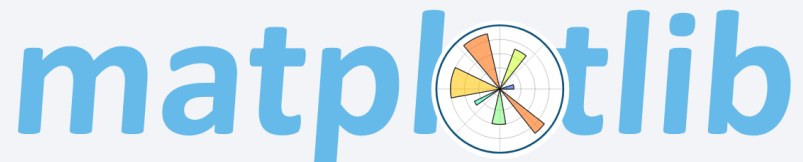
EDA with Data Visualization

Purpose of visualizations

- To identify patterns and relationships between features and launch outcomes.
- To analyze which variables, impact the success rate of Falcon 9 booster landings.
- To support feature selection for training supervised classification models.

Charts used (using Matplotlib and Seaborn libraries)

- Bar charts: used to visualize the relationship between categorical variables (e.g., orbit type) and success rate.
- Scatterplots: used to analyze the relationship between two numerical variables (e.g., flight number and payload mass) or a numerical variable and a categorical variable (e.g., payload mass and launch site).
- Line plots: used to observe trends in landing success over time.



EDA with SQL



Purpose

- To analyze relationships between categorical variables and landing success.
- To validate findings from exploratory data analysis using structured SQL queries.
- To identify key features influencing Falcon 9 booster landing outcomes.

Analyses

- Queried distinct launch sites and filtered missions using string patterns.
- Used aggregate functions (SUM, AVG, COUNT) to analyze payload mass and mission outcomes.
- Identified key events such as the first successful landing and maximum payload boosters using subqueries.
- Analyzed time- and outcome-based patterns by ranking landing outcomes and extracting date-specific records.

Build an Interactive Map with Folium

Purpose of visualizations

- To visualize the geographical distribution of Falcon 9 launch sites and landing outcomes.
- To analyze the impact of launch site location and proximity to infrastructure on landing success.

Markers used and reasoning

- Markers: used to indicate the exact locations of launch sites and individual launch events.
- Circles: used to highlight launch site areas and improve visibility of site locations.
- Marker clusters: used to group multiple launch records at the same coordinates and reduce map clutter.
- Color-coded markers: used to distinguish successful launches from failed launches.
- Polylines: used to visualize distances between launch sites and nearby features such as coastlines, cities, railways, and highways.
- MousePosition tool: used to interactively retrieve latitude and longitude values for distance calculation.

Build a Dashboard with Plotly Dash

Purpose of dashboard visualizations

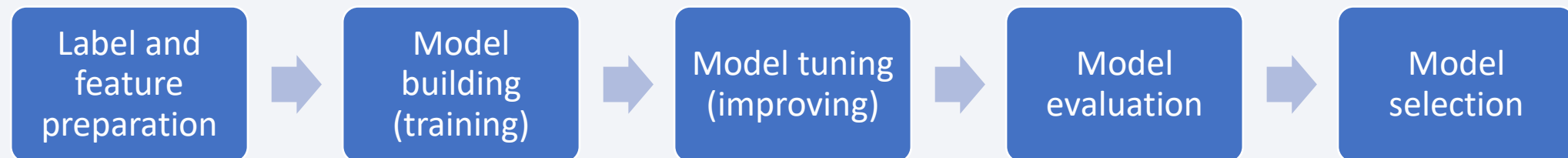
- To interactively explore the relationship between launch site, payload mass, and landing success.
- To identify patterns and correlations that influence Falcon 9 booster landing outcomes and support model feature selection.

Plots and interactions used

- Dropdown menu: used to filter all visualizations by launch site or view results for all sites.
- Pie chart: used to display total successful launches by site, or success vs. failure distribution for a selected site.
- Range slider: used to filter launches by payload mass range.
- Scatter plot: used to analyze the relationship between payload mass and launch success.
- Callbacks: used to dynamically update plots based on user selections and interactions.

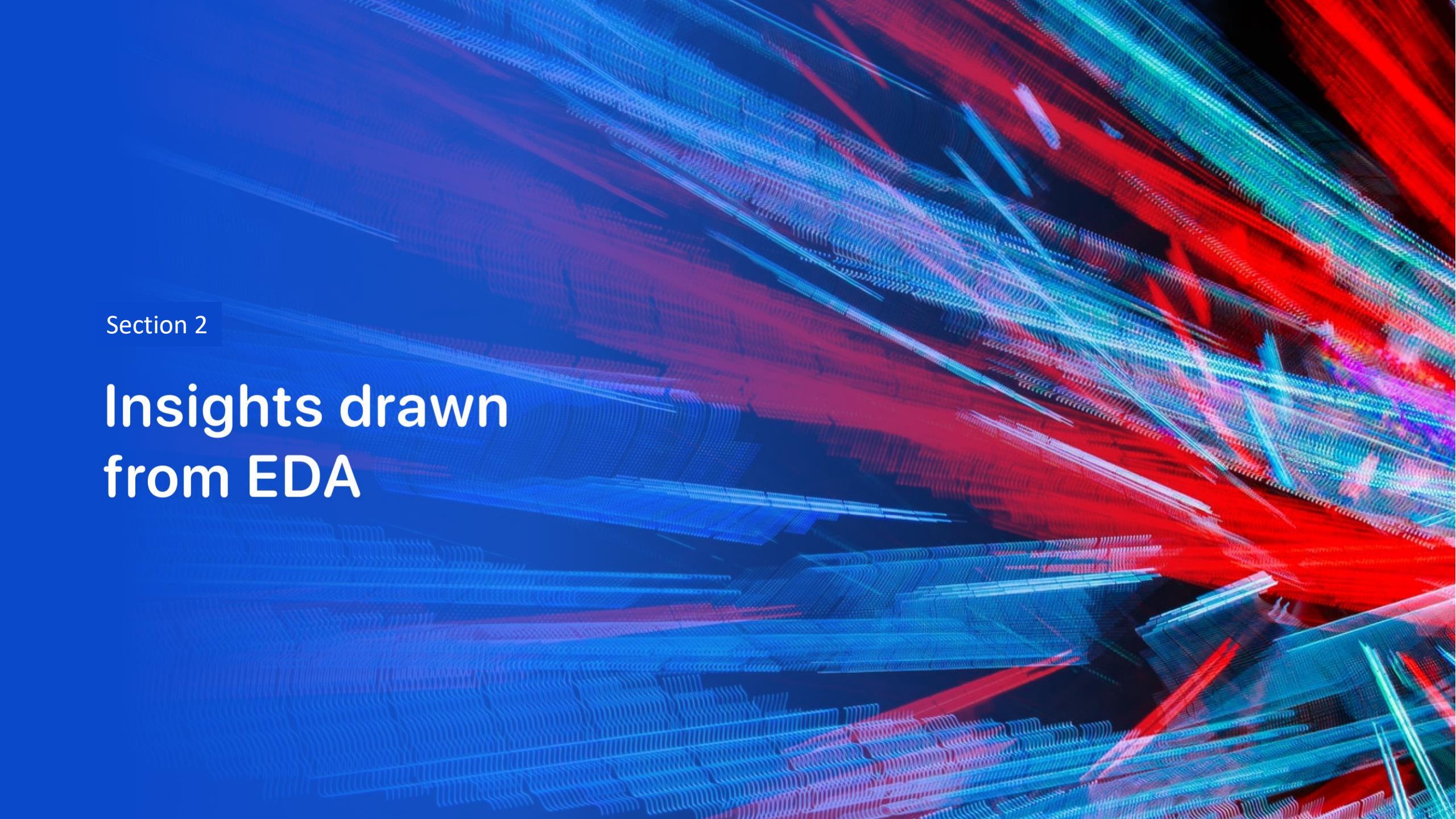
Predictive Analysis (Classification)

- **Label and feature preparation:** created binary target label (Class), standardized numerical features, and split data into training and test sets.
- **Model building:** trained multiple classification models including Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- **Model improvement:** tuned hyperparameters using cross-validation.
- **Model evaluation:** evaluated models using accuracy scores and confusion matrices.
- **Model selection:** selected the best-performing model based on validation and test accuracy.



Results

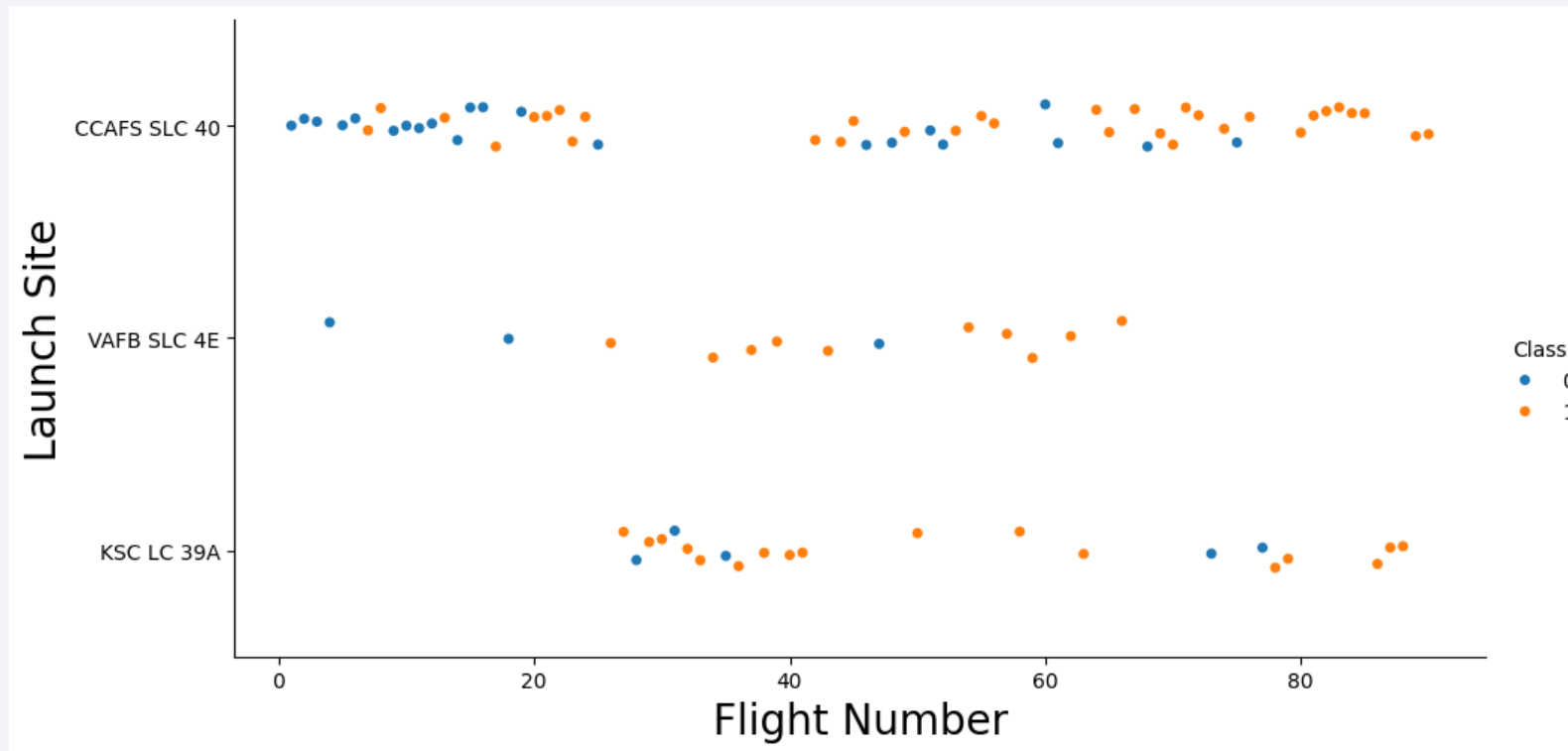
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

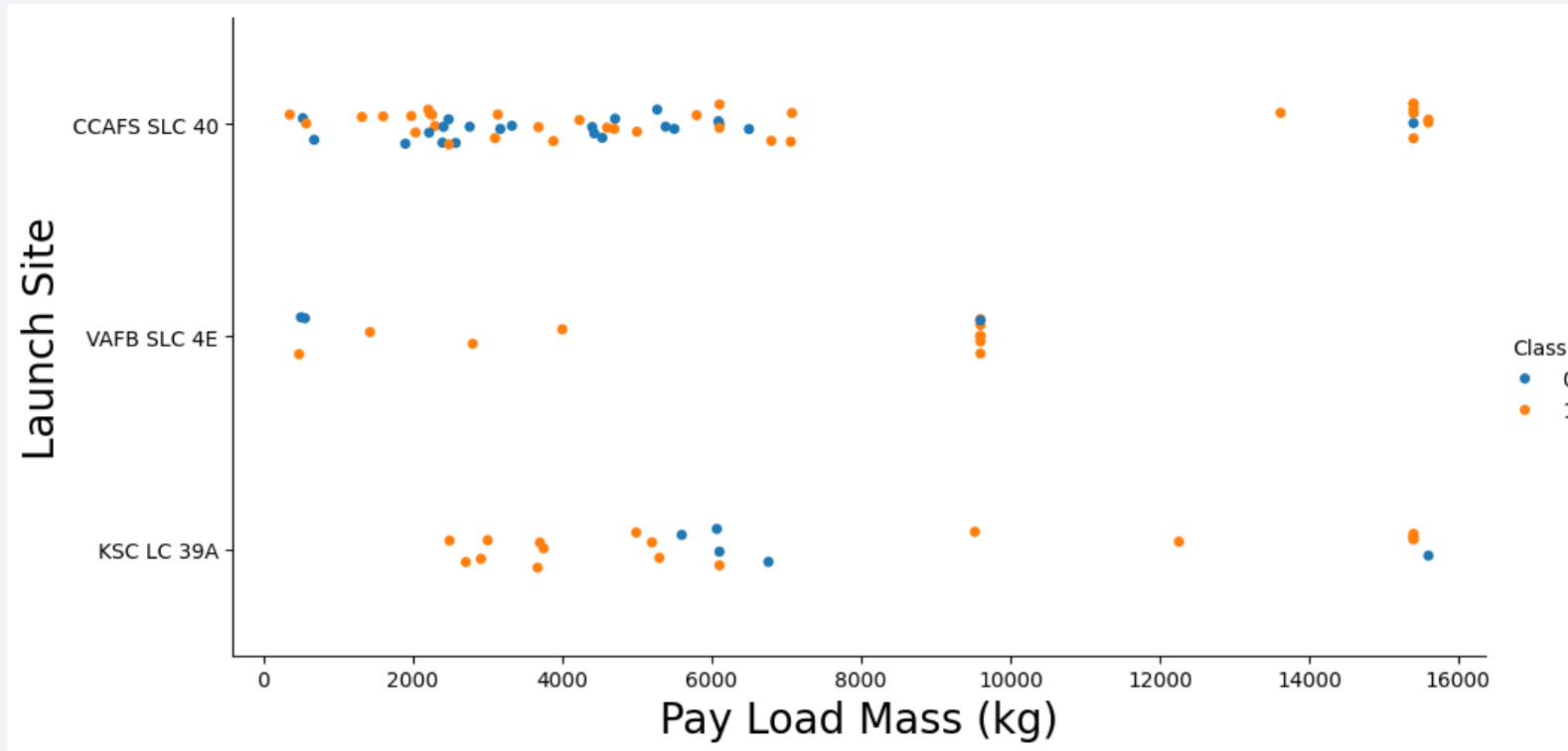
Insights drawn from EDA

Flight Number vs. Launch Site



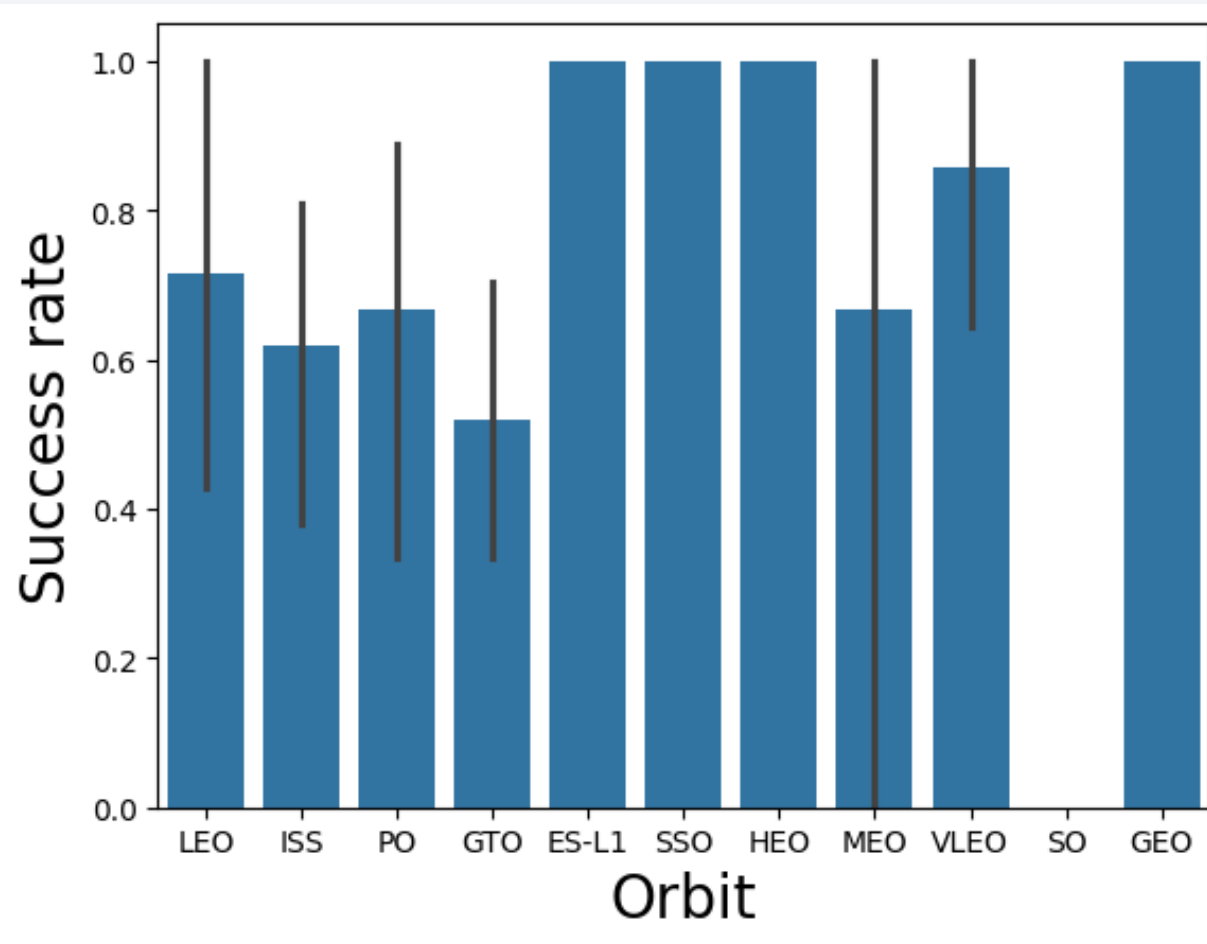
- **Landing success increases over time:** as flight number increases, the proportion of successful landings (Class = 1) rises across all launch sites.
- **Site-dependent performance:** CCSFS SLC-40 and KSC LC-39A show higher frequencies of successful landings compared to VAFB SLC-4E, especially in later flights

Payload vs. Launch Site



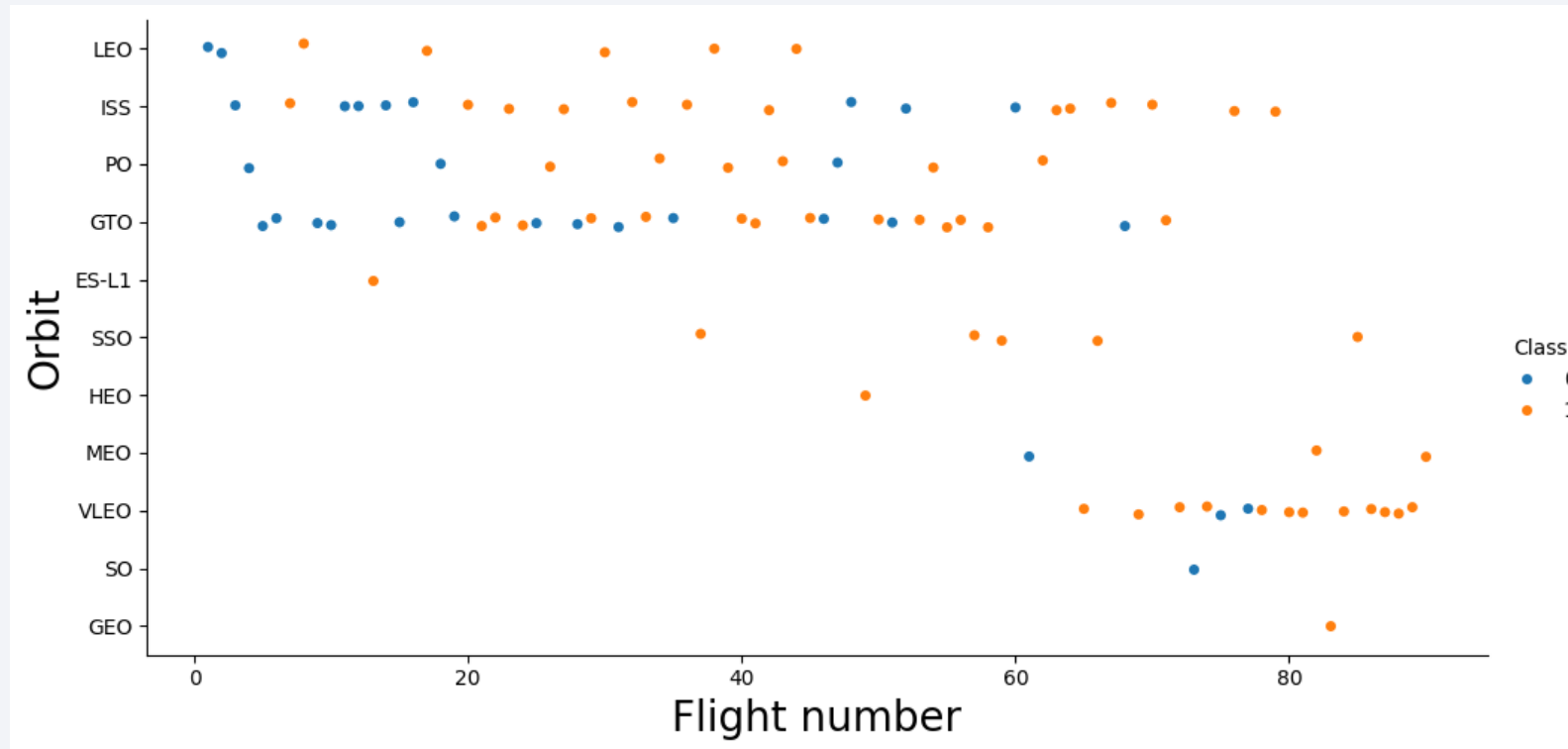
- **Payload mass influences success:** successful landings (Class = 1) occur across a wide range of payload masses, but higher payloads tend to show more successful landings.
- **Launch-site differences:** CCSFS SLC-40 and KSC LC-39A support successful landings over a broader payload mass range compared to VAFB SLC-4E (no payloads > 10000 kg), indicating site-dependent performance characteristics.

Success Rate vs. Orbit Type



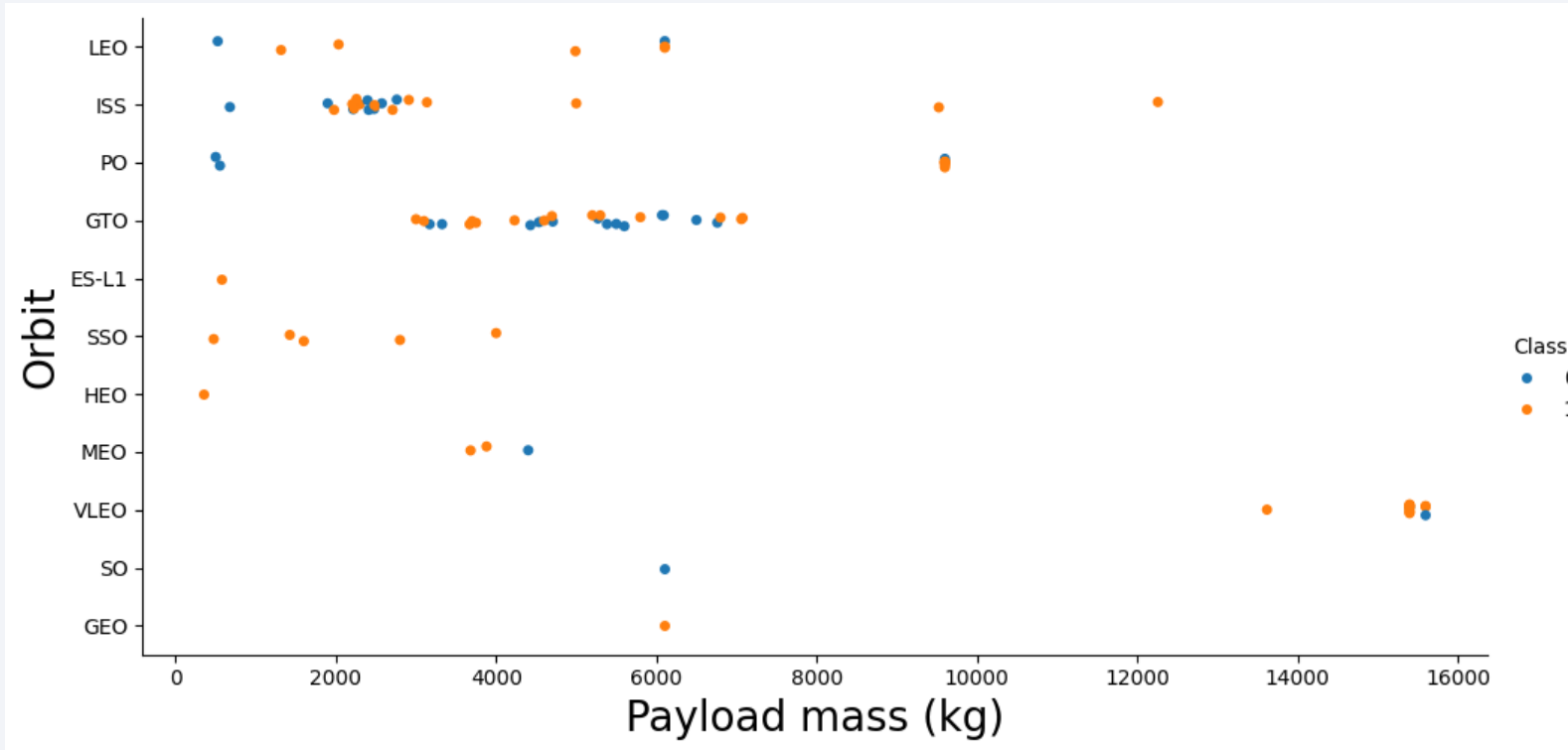
- **Consistent success:** orbits such as ES-L1, SSO, and GEO exhibit consistently high success rates, suggesting stable mission profiles and well-established operational procedures.
- **Greater variability:** orbits like LEO, ISS, PO, and GTO show lower and more variable success rates, indicating greater operational complexity and higher landing risk for these mission types.

Flight Number vs. Orbit Type



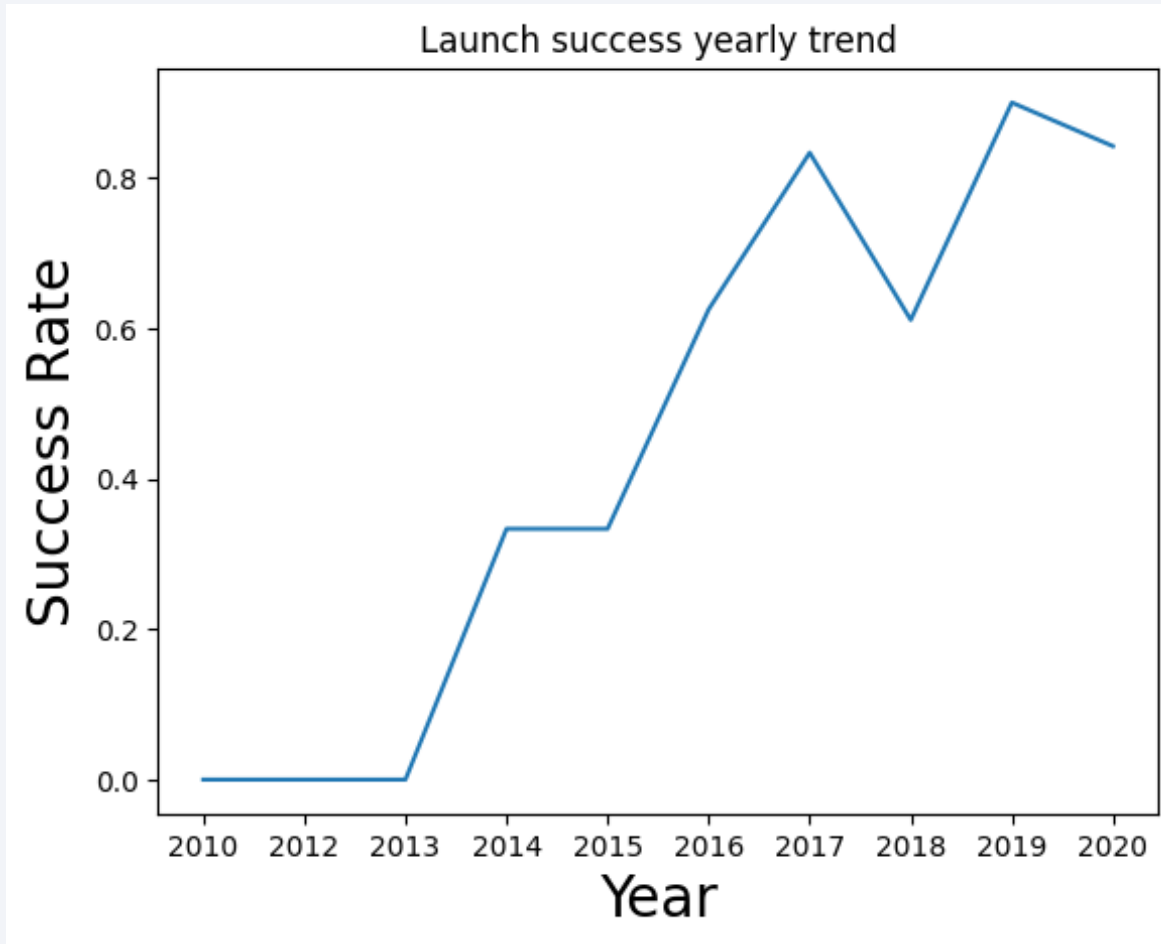
- **Improvement over time:** as the flight number increases, the frequency of successful landings (Class = 1) increases for most orbit types, indicating operational improvement over successive missions.
- **Orbit-dependent patterns:** earlier flights to LEO, ISS, PO, and GTO show more failures, while later missions (especially to VLEO, SSO, and GEO) are dominated by successful landings, suggesting higher maturity in these mission profiles.

Payload vs. Orbit Type



- **Orbit-specific payload regimes and success patterns:** higher payload masses are more common in orbits such as VLEO, while LEO, ISS, and PO missions typically carry lower to mid-range payloads but show higher successful landing rates even with heavier payloads.
- **Payload mass alone is not decisive:** while successful landings occur across a wide range of payload masses, GTO missions exhibit both successful and unsuccessful outcomes at similar payload levels, indicating that landing success depends on orbit characteristics in addition to payload mass.

Launch Success Yearly Trend



- **Clear improvement over time:** launch success rates increase significantly after 2013, indicating rapid operational learning and maturation of Falcon 9 landing capabilities.

All Launch Site Names

Query used

```
%%sql
```

```
SELECT DISTINCT
```

```
LAUNCH_SITE AS Launch_Sites
```

```
FROM SPACEXTBL;
```

Remove duplicates

Find the launch sites

Source

Output

Launch_Sites

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Query used

```
%%sql
SELECT
  *
FROM SPACEXTBL
WHERE
  LAUNCH_SITE LIKE 'CCA%' -- matches sites beginning with 'CCA'
ORDER BY
  DATE
LIMIT 5;
```

Output

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Filters launch records to include only those from launch sites whose names begin with “CCA”.
- The ORDER BY DATE clause sorts the results in chronological order.
- The LIMIT 5 clause restricts the output to the first five launch records.

Total Payload Mass

Query used

```
%%sql
SELECT
  SUM(PAYLOAD_MASS_KG_) AS total_payload_mass_nasa_crs
FROM SPACEXTBL
WHERE
  CUSTOMER = 'NASA (CRS)';
```

Output

total_payload_mass_nasa_crs
45596

- Filters launch records to include only those where the customer was NASA (CRS).
- The SUM function calculate the total payload mass across all missions from NASA (CRS).

Average Payload Mass by F9 v1.1

Query used

```
%%sql
SELECT
  AVG(PAYLOAD_MASS_KG_) AS avg_payload_mass_f9_v1_1
FROM SPACEXTBL
WHERE
  BOOSTER_VERSION = 'F9 v1.1';
```

Output

avg_payload_mass_f9_v1_1
2928.4

- Filters launch records to include only those where the booster version was F9 v1.1.
- The AVG function calculate the average payload mass across all missions where the booster version is F9 v1.1 was used

First Successful Ground Landing Date

Query used

```
%%sql
SELECT
  MIN(DATE) AS date_first_successful_landing_in_ground_pad
FROM SPACEXTBL
WHERE
  LANDING_OUTCOME = 'Success (ground pad)';
```

Output

date_first_successful_landing_in_ground_pad
2015-12-22

- Filters launch records to include only missions with a successful round-pad landing.
- The MIN(DATE) function identifies the earliest date on which a successful round-pad landing was achieved.

Successful Drone Ship Landing with Payload between 4000 and 6000

Query used

```
%%sql
SELECT
  BOOSTER_VERSION
FROM SPACEXTBL
WHERE
  LANDING_OUTCOME = 'Success (drone ship)'
  AND PAYLOAD_MASS_KG_ >= 4000
  AND PAYLOAD_MASS_KG_ <= 6000;
```

- Filters launch records to include only missions with a successful landing on a drone ship and with payload mass between 4000 and 6000 kg.

Output

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Query used

```
%%sql
SELECT
    success_table.num_success_outcomes,
    failure_table.num_failure_outcomes
FROM
    (SELECT COUNT(*) AS num_success_outcomes
     FROM SPACEXTBL
     WHERE MISSION_OUTCOME LIKE 'Success%') AS success_table,
    (SELECT COUNT(*) AS num_failure_outcomes
     FROM SPACEXTBL
     WHERE MISSION_OUTCOME LIKE 'Failure%') AS failure_table;
```

Output

num_success_outcomes	num_failure_outcomes
100	1

- Uses two subqueries to independently count the successful and failed missions.
- The LIKE 'Success%' and LIKE 'Failure%' ensures all variations of success and failure are included

Boosters Carried Maximum Payload

Query used

```
%%sql
SELECT DISTINCT
  BOOSTER_VERSION
FROM SPACEXTBL
WHERE PAYLOAD_MASS_KG = (
  SELECT MAX(PAYLOAD_MASS_KG_)
  FROM SPACEXTBL
);
```

Output

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- Uses a subquery to determine the maximum payload mass in the dataset.
- Uses DISTINCT to select all unique booster versions that carried this maximum payload mass.

2015 Launch Records

Query used

```
%%sql
SELECT
  CASE
    WHEN substr(DATE, 6, 2) = '01' THEN 'January'
    WHEN substr(DATE, 6, 2) = '02' THEN 'February'
    WHEN substr(DATE, 6, 2) = '03' THEN 'March'
    WHEN substr(DATE, 6, 2) = '04' THEN 'April'
    WHEN substr(DATE, 6, 2) = '05' THEN 'May'
    WHEN substr(DATE, 6, 2) = '06' THEN 'June'
    WHEN substr(DATE, 6, 2) = '07' THEN 'July'
    WHEN substr(DATE, 6, 2) = '08' THEN 'August'
    WHEN substr(DATE, 6, 2) = '09' THEN 'September'
    WHEN substr(DATE, 6, 2) = '10' THEN 'October'
    WHEN substr(DATE, 6, 2) = '11' THEN 'November'
    WHEN substr(DATE, 6, 2) = '12' THEN 'December'
  END AS month_name,
  LANDING_OUTCOME,
  BOOSTER_VERSION,
  LAUNCH_SITE
FROM SPACEXTBL
WHERE
  substr(DATE, 1, 4) = '2015'
  AND LANDING_OUTCOME LIKE 'Failure (drone ship)%'
ORDER BY
  substr(DATE, 6, 2);
```

Output

month_name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Filters launch records from 2015, restricts the results to failed drone-ship landing attempts (LANDING_OUTCOME LIKE 'Failure (drone ship)%'), and sort results in chronological order.
- Uses CASE statement to convert numeric month values into month names.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query used

```
%%sql
SELECT
    ... LANDING_OUTCOME,
    ... COUNT(*) AS Landing_Count
FROM SPACEXTBL
WHERE
    ... DATE >= '2010-06-04'
    ... AND DATE <= '2017-03-20'
GROUP BY
    ... LANDING_OUTCOME
ORDER BY
    ... Landing_Count DESC;
```

Output

Landing_Outcome	Landing_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

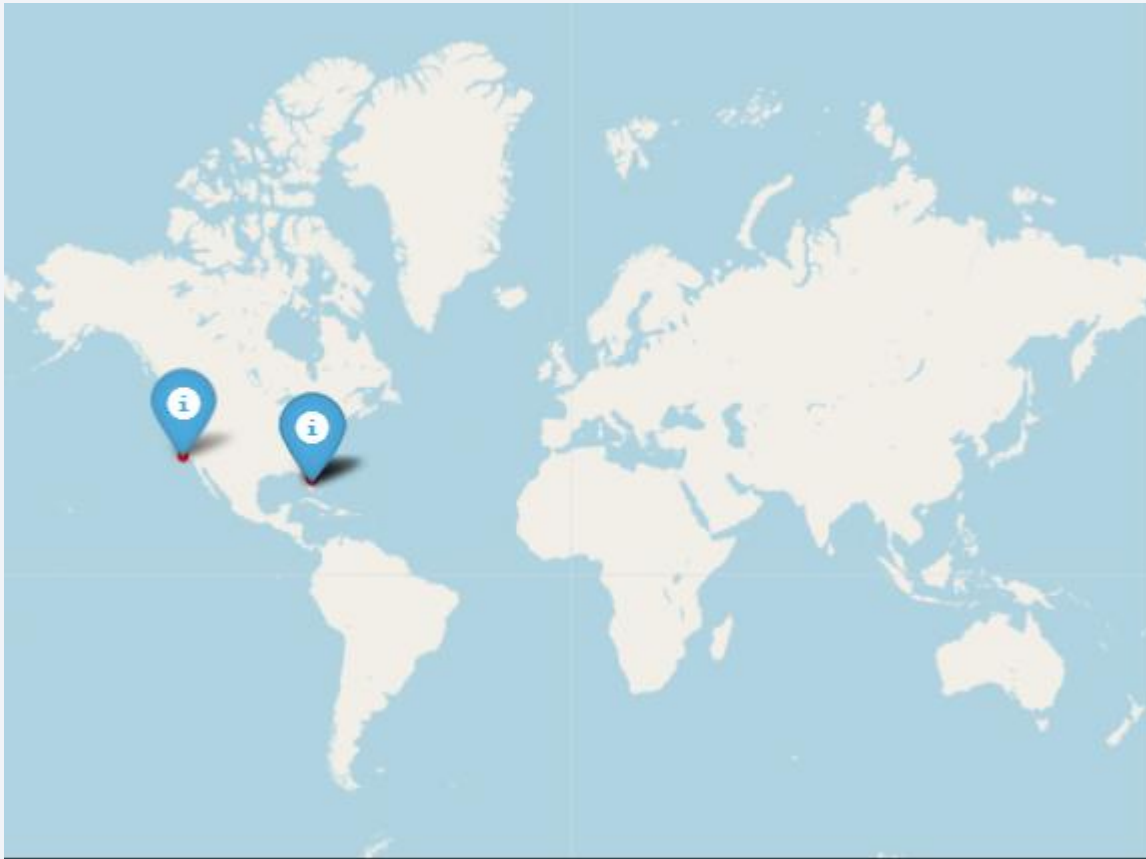
- Filters launch records to include missions that occurred between 2010-07-04 and 2017-03-20, group the data based on the landing outcome, and sort the results in descending (from largest to smallest) order.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Global locations of Falcon 9 Launches



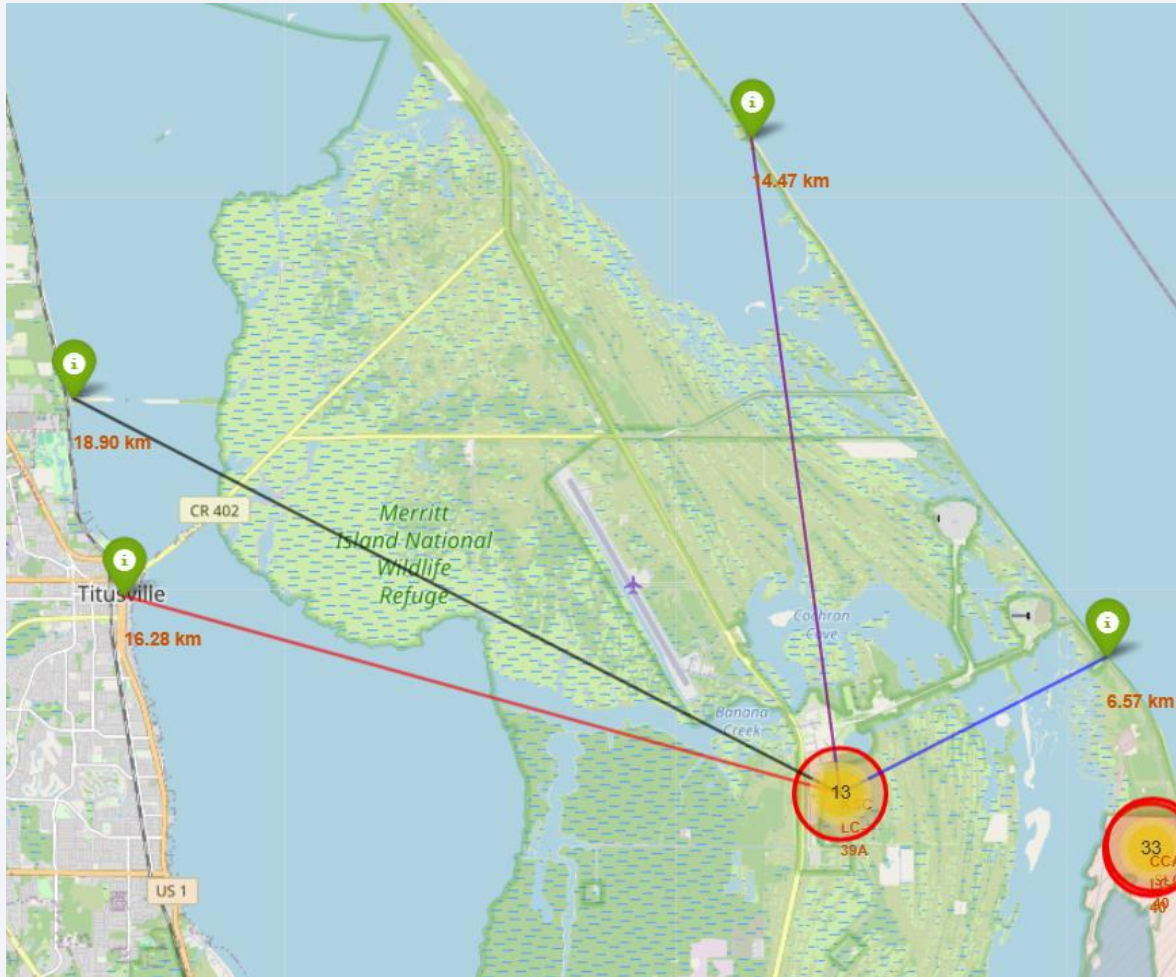
- The map shows distinct markers of the launch locations (e.g., CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E). Some of these locations are close to each other and given the zoom on the image, may have markers that seem to be overlapping.
- All markers/locations are within North America, specifically in the USA
- The launch locations are placed in coastal areas, and in this example, close to the Atlantic and Pacific coasts.

Launch outcomes at Kennedy Space Center

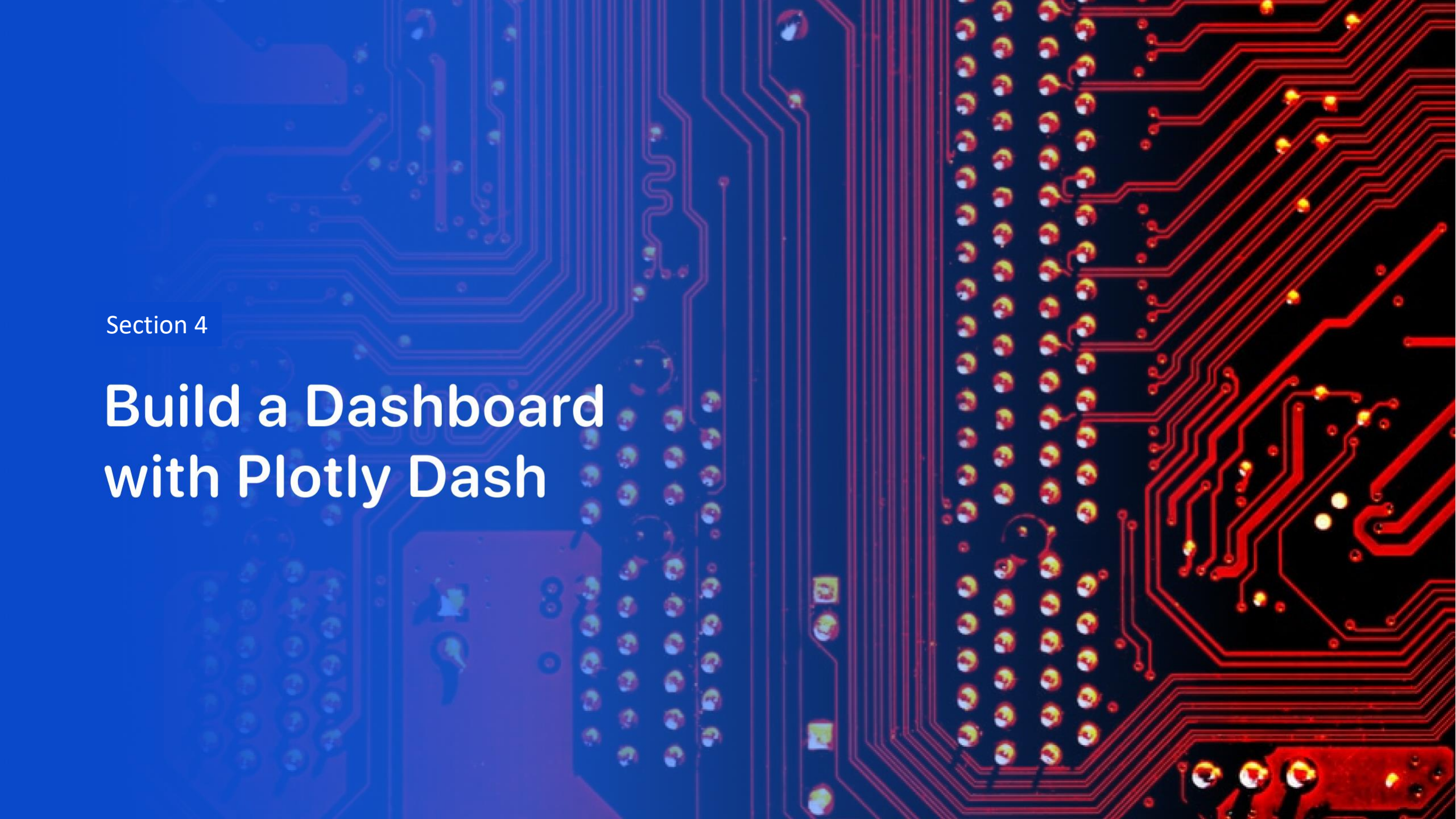


- Each colored marker represents an unique Falcon 9 launch:
 - Green: launches with successful landings
 - Red: launches with failed landings
- Based on the ratio between green and red markers, it can be inferred that KSC LC-39A has a high rate of successful landings

Proximity Analysis of KSC LC-39A



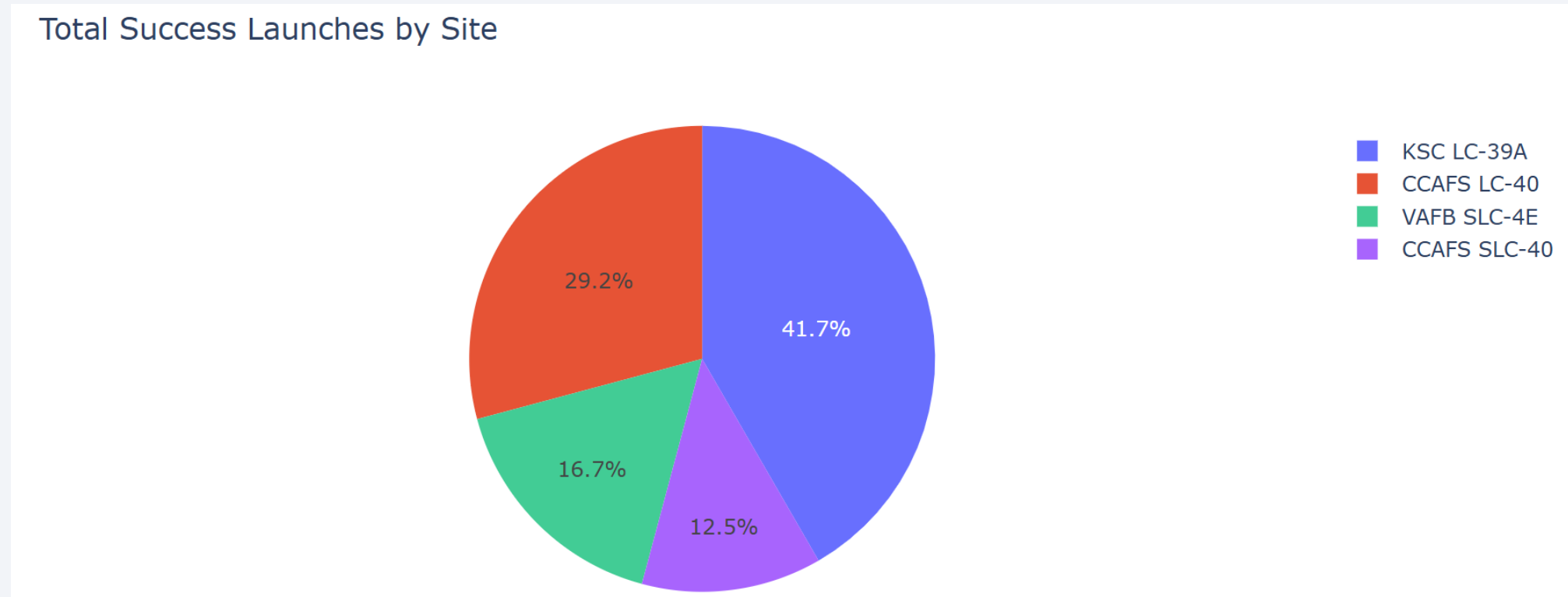
- Markers placed at nearby points of interest show calculated distances (in kilometers) from the launch site to:
 - The coastline – 6.57 km
 - The nearest city (e.g., Titusville) – 16.28 km
 - Major highways (e.g., Playalinda Beach Road (Beach Road/FL-402)) – 14.47 km
 - Railway infrastructure (e.g., Jay Jay Yard) – 18.9 km
- The map clearly indicates the proximity to the coastline and relative separation from urban areas, while having strategic infrastructure in the vicinity



Section 4

Build a Dashboard with Plotly Dash

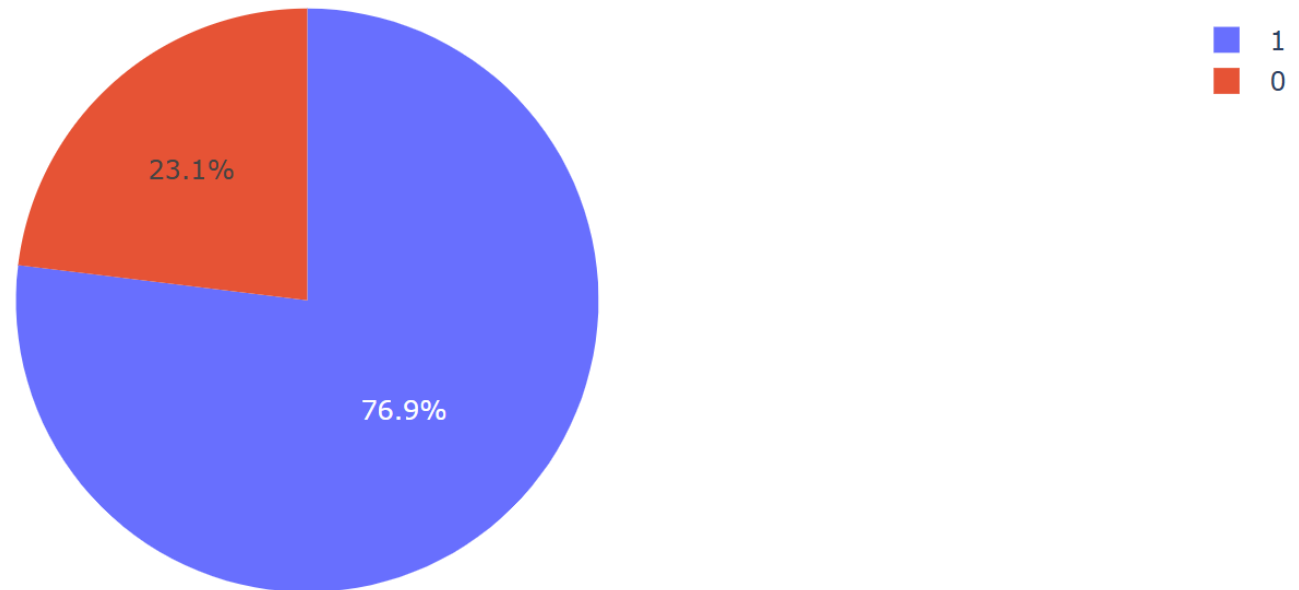
Total success launches by site



- KSC LC-39A accounts for the largest share of successful launches (~41.7%), while CCAFS SLC-40 accounts for the lowest share of successful missions (~12.5 %)

Success vs failure ratio for KSC LC-39A

Success vs Failure for site KSC LC-39A



- Color coded segments: successful (1 - blue) and failure (2 - red)
- The chart indicates that approximately 76.9% of launches were successful, while 23.1% resulted in failure at KSC LC-39A, which indicates site maturity and reliability.

Payload Range 2500–7000 kg

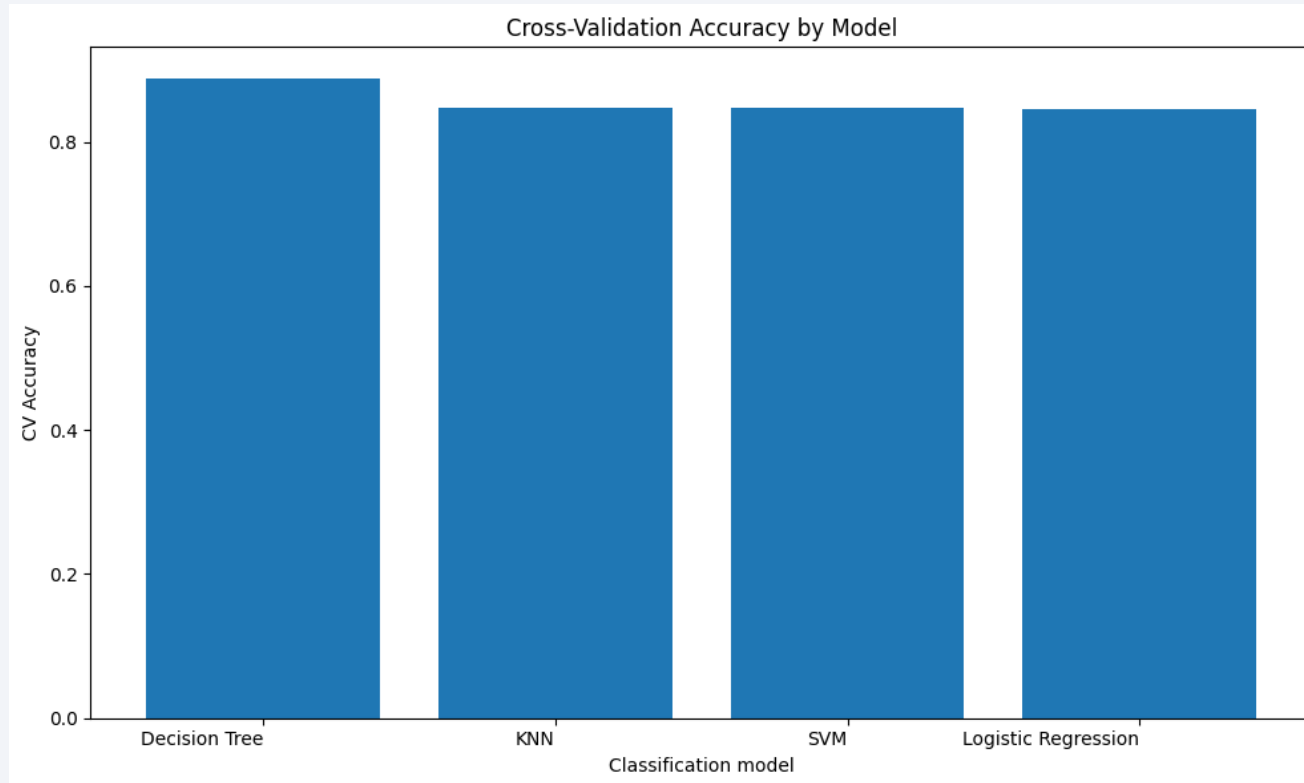


- Scatterplot represents the relationship between Class (success = 1, and failure = 0) and payload mass.
- More successful missions when payload mass is in the 2500-5000 kg range.
- Increased number of failed missions when payload mass >2500 kg.

Section 5

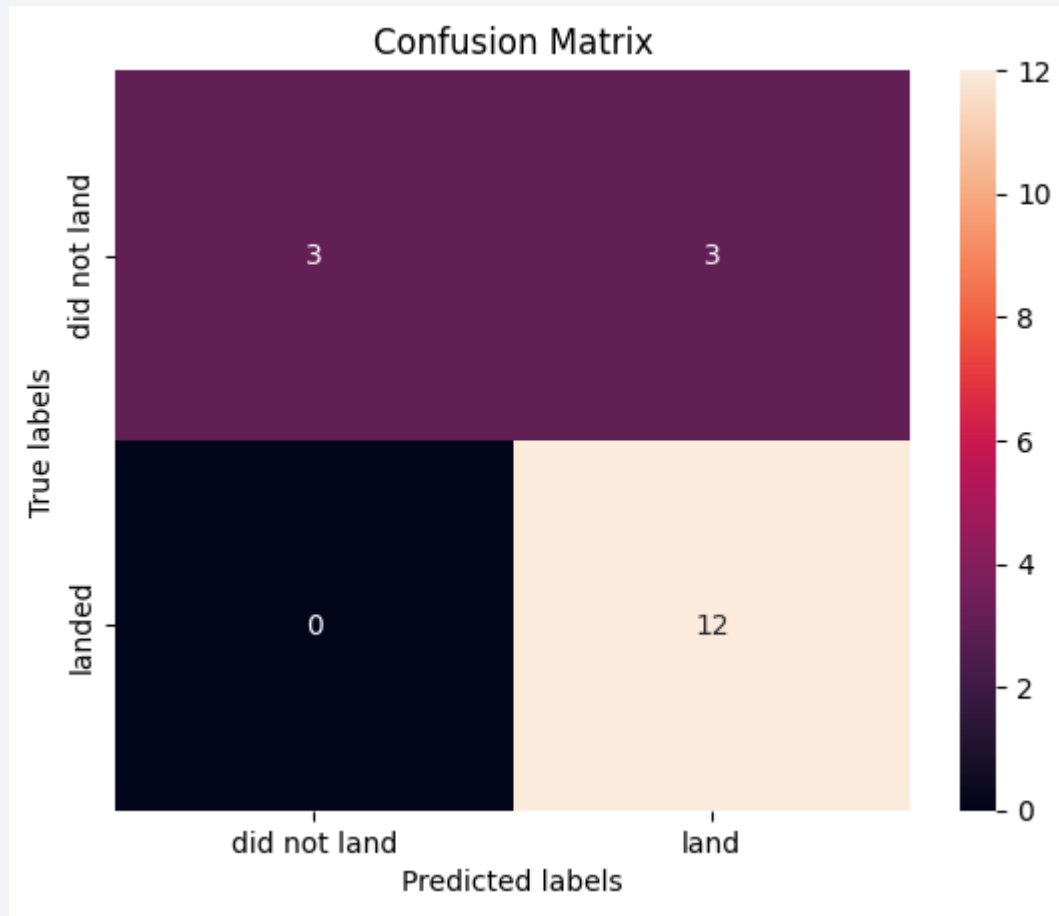
Predictive Analysis (Classification)

Classification Accuracy



- Bar plot displaying the cross-validation accuracy for the different classification models.
- Decision tree exhibited the highest classification accuracy (0.8875) from the four models evaluated.

Confusion Matrix (decision tree model)



- The model correctly predicts all 12 successful landings (True positives) without false negatives (predict that it land when it failed)
- 3 false positives (predicted that it landed when it failed).

Conclusions

- There are many different features that clearly affect the Falcon 9 first-stage landing, with payload mass, orbit type, booster version, and launch site being identified as key features.
- The success rate of landings improved significantly over time, showing improvements in operations, maturity, and increased in reliability.
- Multiple classification models were trained and evaluated, and hyperparameter tuning improved overall predictive performance of the models.
- Decision tree model exhibited the best overall predictive performance (highest accuracy), making the best model (considered in this exercise) for predicting the outcome of landing.

Thank you!

